

Analysis 1
final exam of real analysis

Exercise 1.

Consider the function f defined on $\mathbb{R}$ by $f(x) = \frac{\sin(\pi x) + 1}{x^2 - 2x + 5}$.

1. Justify the differentiability of f on $\mathbb{R}$ and calculate the derivative of the function f .
2. Deduce that the equation

$$(x^2 - 2x + 5)(\pi \cos(\pi x)) - (2x - 2)(1 + \sin(\pi x)) = 0,$$

has at least one solution $c \in]0, 2[$.

Exercise 2.

- 2) Find, if they exist, the infimum and supremum of the following set:

$$C = \left\{ 2 + (-1)^n + \left(\frac{1}{3}\right)^n, \quad n \in \mathbb{N} \right\}.$$

- 3) Let X be a nonempty and bounded subset of $\mathbb{R}$ and $a > 0$, let Y the set defined by $Y = \{ax; x \in X\}$. Show that $\sup Y = a \sup X$.

Exercise 3. We consider the recursive sequence $(u_n)_{n \in \mathbb{N}}$ defined by:

$$\begin{cases} u_0 = a, & a \in \mathbb{R}^+, \\ u_{n+1} = \frac{1}{2}u_n(u_n + 1), & n \in \mathbb{N}. \end{cases}$$

I) Suppose that $a = \frac{1}{2}$.

1. Prove that $\forall n \in \mathbb{N}, 0 < u_n < 1$.
2. Study the monotonicity of the sequence (u_n) .
3. Deduce that (u_n) is convergent and find its limit.

II) Suppose that $a > 1$.

1. Prove that $\forall n \in \mathbb{N}, u_n > 1$.
2. Show that (u_n) is strictly increasing and that $u_n \geq \left(\frac{1+a}{2}\right)^n, \forall n \in \mathbb{N}$.
3. Deduce the nature of the sequence (u_n) .

Exercise 4. 1. Show that:

$$\forall x > 0, \quad \frac{1}{x+1} < \ln(x+1) - \ln(x) < \frac{1}{x}.$$

2. Calculate the following limit:

$$\lim_{x \rightarrow +\infty} x^\alpha \times (\ln(x+1) - \ln(x)), \quad \alpha \geq 1.$$

4. Deduce the following limit:

$$\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right)^x.$$

Correction :

Exercise 1 :

1. The function f is differentiable on $\mathbb{R}$ since it is a quotient [0.5pt] of two differentiable functions on $\mathbb{R}$ and for all $x \in \mathbb{R}$ we have $x^2 - 2x + 5 > 0$ [0.5pt]. Using the quotient rule we obtain

$$f'(x) = \frac{(x^2 - 2x + 5)(\pi \cos(\pi x)) - (2x - 2)(1 + \sin(\pi x))}{(x^2 - 2x + 5)^2}. [0.5pt]$$

2. The function f is differentiable and continuous on the interval $]0, 2[$ [0.5pt], furthermore $f(0) = f(2) = \frac{1}{5}$ [0.5pt], we apply Roll's theorem [0.5pt] and we deduce that there exist $c \in]0, 2[$ such that $f'(c) = 0$. [0.5pt] Hence

$$(c^2 - 2c + 5)(\pi \cos(\pi c)) - (2c - 2)(1 + \sin(\pi c)) = 0.$$

Exercise 2 :

1. Remark that $(-1)^n$ alternates between 1 when n is even and -1 when n is odd, then we can write $C = C_1 \cup C_2$ with $C_1 = \{3 + \frac{1}{3^{2n}}, n \in \mathbb{N}\}$ [0.5pt] and $C_2 = \{1 + \frac{1}{3^{2n+1}}, n \in \mathbb{N}\}$ [0.5pt].

Now, we denote by $u_n = 3 + \frac{1}{3^{2n}}$ and $v_n = 1 + \frac{1}{3^{2n+1}}, n \in \mathbb{N}$ its not difficult to see that the sequences (u_n) and (v_n) are increasing [0.5pt] which leads to deduce that

$$\inf C_1 = \lim_{n \rightarrow +\infty} u_n = 3 [0.25pt], \inf C_2 = \lim_{n \rightarrow +\infty} v_n = 1, [0.25pt]$$

$$\sup C_1 = u_0 = 4 [0.25pt], \sup C_2 = v_0 = \frac{4}{3}. [0.25pt]$$

Finally we conclude that $\sup C = \max\{\sup C_1, \sup C_2\} = 4$ [0.25pt] and $\inf C = \min\{\inf C_1, \inf C_2\} = 1$. [0.25pt]

2. By the fact that $a > 0$ we have

$$\forall x \in X : x \leq \sup(X) \implies \forall x \in X : ax \leq a \sup(X).$$

So:

$$\forall y \in Y : y = ax \leq a \sup(X). [0.5pt]$$

This implies that $a \sup(X)$ is an upper bound of Y , and as $\sup(Y)$ is the smallest upper bound of Y , [0.5pt] then:

$$\sup(Y) \leq a \sup(X). [0.5pt]$$

On the other hand:

$$\forall x \in X : ax \leq \sup(Y) \iff \forall x \in X : x \leq \frac{\sup(Y)}{a}. [0.5pt]$$

This implies that $\frac{\sup(Y)}{a}$ is an upper bound of X , and as $\sup(X)$ is the smallest upper bound of X , then: [0.5pt]

$$\sup(X) \leq \frac{\sup(Y)}{a}.$$

So:

$$a \sup(X) \leq \sup(Y).$$

[0.5pt] We have:

$$\sup(Y) \leq a \sup(X) \quad \text{and} \quad a \sup(X) \leq \sup(Y) \implies \sup(Y) = a \sup(X).$$

Exercise 3:

I) Suppose that $a = \frac{1}{2}$.

1. Prove that $\forall n \in \mathbb{N}, 0 < u_n < 1$.

Base case ($n = 0$): By definition, $u_0 = \frac{1}{2}$, so $0 < u_0 < 1$ [0.5pt].

Inductive step: Assume that $0 < u_n < 1$ holds for some $n \in \mathbb{N}$. We need to prove that $0 < u_{n+1} < 1$. Since $u_n > 0$, we have $u_n + 1 > 1$, its clear that

$$u_{n+1} = \frac{1}{2}u_n(u_n + 1) > 0. \text{[0.5pt]}$$

Moreover Since $0 < u_n < 1$, we have $u_n + 1 < 2$. Thus:

$$u_{n+1} = \frac{1}{2}u_n(u_n + 1) < \frac{1}{2} \cdot 1 \cdot 2 = 1.$$

Hence by induction, $\forall n \in \mathbb{N}$, $0 < u_n < 1$. [0.5pt]

2. We study the monotonicity, of the sequence (u_n) :

$$u_{n+1} - u_n = \frac{1}{2}u_n(u_n + 1) - u_n = u_n \left(\frac{1}{2}(u_n + 1) - 1 \right) = u_n \times \frac{u_n - 1}{2}. \text{[0.5pt]}$$

Since $0 < u_n < 1$, then $u_n - 1 < 0$, so $u_{n+1} - u_n < 0$. Thus, (u_n) is decreasing [0.5pt].

3. Deduce that (u_n) is convergent and find its limit. Since (u_n) is bounded below by 0 (as $u_n > 0$) and is decreasing, it converges by the monotone convergence theorem [0.5pt]. Let L be the limit of (u_n) . we pass to the limit in the recurrence relation:

$$L = \frac{1}{2}L(L + 1) \implies L(L - 1) = 0. \text{[0.5pt]}$$

Thus, $L = 0$ or $L = 1$. Since $0 < u_n < 1$, we conclude that $L = 0$. [0.5pt]

II) Suppose that $a > 1$: we prove by induction that $\forall n \in \mathbb{N}$, $u_n > 1$.

Base case ($n = 0$): By definition, $u_0 = a > 1$, so $u_0 > 1$ [0.25pt].

Inductive step: Assume that $u_n > 1$ for some $n \in \mathbb{N}$. Since $u_n > 1$, we have $u_n + 1 > 2$. Thus:

$$u_{n+1} > \frac{1}{2} \times 1 \times 2 = 1.$$

By induction, $\forall n \in \mathbb{N}$, $u_n > 1$ [0.5pt].

2. Show that (u_n) is strictly increasing and $u_n \geq \left(\frac{1+a}{2}\right)^n$, $\forall n \in \mathbb{N}$, we have

$$u_{n+1} - u_n = u_n \times \frac{u_n - 1}{2}. \text{[0.25pt]}$$

Since $u_n > 1$, we have $u_n - 1 > 0$, so $u_{n+1} - u_n > 0$. Thus, (u_n) is strictly increasing [0.5pt]. Now, prove that $u_n \geq \left(\frac{1+a}{2}\right)^n$ by induction:

Base case ($n = 0$):

$$u_0 = a \geq \left(\frac{1+a}{2}\right)^0 = 1. \text{[0.25pt]}$$

Inductive step: Assume $u_n \geq \left(\frac{1+a}{2}\right)^n$ for some $n \in \mathbb{N}$. Using the inductive hypothesis, $u_n \geq \left(\frac{1+a}{2}\right)^n$, and since $u_n > u_0 = a$, we get

$$u_{n+1} \geq \left(\frac{1+a}{2}\right)^{n+1}.$$

By induction, $u_n \geq \left(\frac{1+a}{2}\right)^n$, $\forall n \in \mathbb{N}$. [0.5pt]

3. Deduce the nature of the sequence (u_n) . The fact that $a > 1$ gives $\lim_{n \rightarrow +\infty} u_n = +\infty$. [0.5pt] then, the sequence (u_n) is divergent [0.25pt].

Exercise 4 :

1. We prove that

$$\forall x > 0, \quad \frac{1}{x+1} < \ln(x+1) - \ln(x) < \frac{1}{x}.$$

For every $x > 0$ the function $f(t) = \ln(t)$ is continuous and differentiable on the interval $[x, x + 1]$, [0.5pt] by the mean value theorem there exists $c \in (x, x + 1)$ such that:

$$\ln(x + 1) - \ln(x) = f'(c) = \frac{1}{c}. [0.5pt]$$

Since $c \in (x, x + 1)$, it follows that:

$$x < c < x + 1 \implies \frac{1}{x + 1} < \frac{1}{c} < \frac{1}{x}. [0.5pt]$$

Thus:

$$\frac{1}{x + 1} < \ln(x + 1) - \ln(x) < \frac{1}{x}.$$

2. Calculate the limit

$$\lim_{x \rightarrow +\infty} x^\alpha \times (\ln(x + 1) - \ln(x)).$$

From part (1):

$$\frac{1}{x + 1} < \ln(x + 1) - \ln(x) < \frac{1}{x}.$$

Multiply through by x^α :

$$\frac{x^\alpha}{x + 1} < x^\alpha \times (\ln(x + 1) - \ln(x)) < x^{\alpha-1}. [0.5pt]$$

If $\alpha = 1$, as $x \rightarrow +\infty$, $\frac{x}{x+1} \rightarrow 1$. By the squeeze theorem:

$$\lim_{x \rightarrow +\infty} x \times (\ln(x + 1) - \ln(x)) = 1. [0.5pt]$$

If $\alpha > 1$, as $x \rightarrow +\infty$, $\frac{x^\alpha}{x+1} \rightarrow \infty$, we deduce that by the squeeze theorem again:

$$\lim_{x \rightarrow +\infty} x \times (\ln(x + 1) - \ln(x)) = \infty. [0.5pt]$$

3. Let $A(x) = \left(1 + \frac{1}{x}\right)^x$. Then we have

$$\ln(A(x)) = x \times \ln\left(1 + \frac{1}{x}\right) = x \times (\ln(x + 1) - \ln(x)), [0.5pt]$$

by the question 2 we have $\lim_{x \rightarrow +\infty} \ln(A(x)) = 1$. [0.5pt]

Thus:

$$\lim_{x \rightarrow +\infty} A(x) = \lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x}\right)^x = e. [0.5pt]$$